Airlines & Data Science

What is Overbooking?

Overbooking happens when an airline sells more tickets for a flight than there are seats on the plane. Airlines do this because they expect that some passengers won't show up. But sometimes, **everyone shows up**, and there aren't enough seats for all the passengers.

Why Do Airlines Overbook?

- Airlines want to **maximize profits**.
- They know that **some people cancel or miss their flights**, so they sell extra tickets.
- But if **everyone shows up**, they have to **"bump"** some passengers, meaning those people are not allowed to board.

What Happens If You Get Bumped?

- You might be asked to **take a later flight**.
- The airline may offer **compensation** like money, food vouchers, or hotel stays.
- In some countries, there are **laws** that protect passengers and require airlines to compensate them.

How Does Overbooking Work?

Airlines use **past data** to predict how many people won't show up.

- They calculate the **No Show Rate** using this data.

- Then they decide how many **extra tickets** they can safely sell.
- This means they might sell **105 tickets for a plane with only 100 seats**, expecting that 5 people won't show up.

What Factors Do Airlines Consider?

Here are the three main things they look at:

- **No Show Rate:** How many people usually don't show up for this flight.
- **Physical Capacity:** The actual number of seats available for passengers (sometimes fewer than total seats due to crew or equipment).
- **Authorized Capacity:** The number of tickets the airline is allowed to sell, which includes the extra ones based on the no show rate.

Data science can play a powerful role in reducing the problems caused by overbooking. Here's a detailed explanation of how it can help:

1. **Better Prediction of No-Show Rates**

Airlines currently use historical averages to estimate how many people won't show up. But data science can do much better by using Machine learning models that analyze:

- Passenger behavior (e.g., frequent flyers vs. first-time travelers)
- Time of day, day of week, and season
- Weather conditions
- Booking and cancellation patterns

Result: More accurate predictions = fewer people bumped.

2. **Real-Time Decision Making**

With real-time data, airlines can adjust overbooking levels on the fly based on current check-in rates and use dynamic models that update predictions as the flight time gets closer.

Result: Smarter, more flexible decisions.

3. **Passenger Prioritization Models**

Data science can help airlines decide who to bump (if needed) in the most fair and efficient way:

Prioritize based on: Loyalty status, Ticket type (refundable vs. non-refundable), Check-in time, Travel urgency (e.g., connecting flights)

Result: Less frustration, more fairness.

4. **Optimizing Compensation Offers**

Airlines often offer vouchers to volunteers. Data science can help predict the lowest amount that will still get enough volunteers.

Result: Save money while keeping passengers happy.

5. **Simulation and Scenario Testing**

Airlines can use data science to simulate different overbooking strategies and see what works best:

- Try different no-show rates
- Test different compensation levels
- Analyze outcomes before applying them in real life

Result: Safer, data-backed decisions.

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